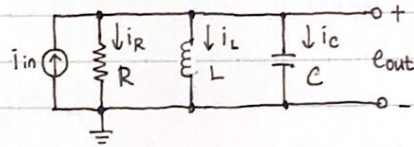


System Modeling - Final Report

Date 2021. 6. 29

Q: Consider the electrical circuit shown in Fig 1, where resistor R , inductor L , and capacitor C are connected parallelly with the current source i_{in} . Consider current i_{in} and the voltage e_{out} as the input and output of the system.



[1] Derive the state space equation for this system.

Elemental law $e_R = R i_R$ $L \frac{di_L}{dt} = e_L$ $C \frac{de_C}{dt} = i_C$

Connection law $e_{out} = e_R = e_L = e_C$
 $i_{in} = i_R + i_L + i_C$

$$e_L = e_R = e_C \Rightarrow L \frac{di_L}{dt} = e_R = e_{out} \Rightarrow \frac{di_L}{dt} = \frac{1}{L} e_{out}$$

$$i_{in} = i_R + i_L + i_C \Rightarrow i_C = i_{in} - i_R - i_L \Rightarrow C \frac{de_C}{dt} = i_{in} - \frac{e_R}{R} - i_L$$

$$\Rightarrow \frac{de_{out}}{dt} = \frac{1}{C} i_{in} - \frac{1}{RC} e_{out} - \frac{1}{C} i_L$$

[2] Write the state space equation in the vector form.

$$\frac{d}{dt} \begin{bmatrix} i_L \\ e_{out} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ e_{out} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{C} \end{bmatrix} i_{in}$$

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[3] Obtain the input-output differential equation relating the input i_{in} to the output e_{out}

Applying Kirchhoff's current law, we get

$$i_c + i_R + i_L - i_{in}(t) = 0$$

Elemental laws:

$$i_R = \frac{1}{R} e_{out}, \quad i_c = C \frac{de_{out}}{dt}, \quad i_L = \underbrace{i_L(0)}_{\text{initial condition for } L} + \frac{1}{L} \int_0^t e_{out}(\lambda) d\lambda.$$

Substituting the elemental laws into Kirchhoff's current law, we get,

$$C \frac{de_{out}}{dt} + \frac{1}{R} e_{out} + \frac{1}{L} \int_0^t e_{out}(\lambda) d\lambda = i_{in}(t) - i_L(0).$$

Differentiate every term, we get

$$C \frac{d^2 e_{out}}{dt^2} + \frac{1}{R} \frac{de_{out}}{dt} + \frac{1}{L} e_{out} = \frac{di_{in}}{dt} - \cancel{\frac{di_L}{dt}} \quad \text{o. (constant)}.$$

$$\Rightarrow C \frac{d^2 e_{out}}{dt^2} + \frac{1}{R} \frac{de_{out}}{dt} + \frac{1}{L} e_{out} = \frac{di_{in}}{dt}$$

$$\text{since } \frac{de_{out}}{dt} = \dot{e}_{out} \text{ and } \frac{d^2 e_{out}}{dt^2} = \ddot{e}_{out},$$

$$\boxed{\frac{di_{in}}{dt} = C \ddot{e}_{out} + \frac{1}{R} \dot{e}_{out} + \frac{1}{L} e_{out}}$$

[4] Write the transfer function $E(s)/I(s)$ from the input i_{in} to the output e_{out} , where $E(s)$ and $I(s)$ are the Laplace-transform of i_{in} and e_{out} with zero initial conditions.

Laplace - transform.

$$L(f(t)) = F(s)$$

$$L(f'(t)) = sF(s) - f(0)$$

$$L(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

For $E(s)$, which is the Laplace transform of LHS ($C\ddot{e}_{out} + \frac{1}{R}\dot{e}_{out} + \frac{1}{L}e_{out}$) would be,

$$C(s^2E(s) - se(0) - e'(0)) + \frac{1}{R}(sE(s) - e(0)) + \frac{1}{L}E(s)$$

Since the initial condition is zero,

$$Cs^2E(s) + \frac{1}{R}sE(s) + \frac{1}{L}E(s) = E(s)(Cs^2 + \frac{1}{R}s + \frac{1}{L})$$

For $I(s)$ is simply

$$L\left(\frac{di_{in}}{dt}\right) = sI(s) - i(0) = sI(s)$$

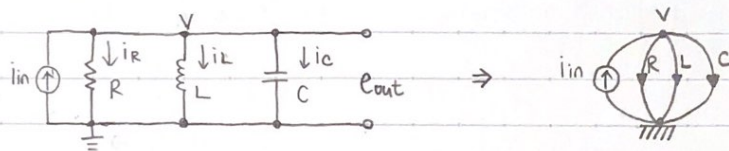
Therefore

$$\frac{E(s)}{I(s)} = \frac{s}{(Cs^2 + \frac{1}{R}s + \frac{1}{L})} = \frac{1}{(Cs + \frac{1}{R} + \frac{1}{LS})} = \boxed{\frac{1}{Cs + R + LS}}$$

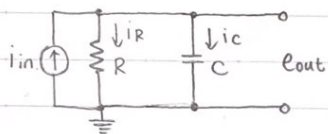
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[5] Draw the linear graph to represent this system.



[6] Consider the case when the inductor L is zero. Obtain the state space equation of the system. Calculate the steady state and time constant of the output response when the input is a unit step ($i_{in}(t) = 1A$ for $t \geq 0$) when $R = 1\Omega$ and $C = 1F$.



Elemental laws $e_R = R i_R$ $C \frac{de_C}{dt} = i_C$

Connection laws. $e_{out} = e_R = e_C$

$i_{in} = i_R + i_C \Rightarrow i_C = i_{in} - i_R$

$$\Rightarrow C \frac{de_C}{dt} = C \frac{de_{out}}{dt} = i_C \Rightarrow \frac{de_{out}}{dt} = \frac{1}{C} i_{in} - \frac{1}{RC} e_{out}$$

$$e_R = e_{out} = R i_R \Rightarrow i_R = \frac{1}{R} e_{out} \Rightarrow \frac{di_R}{dt} = \frac{1}{R} \frac{de_{out}}{dt} = \frac{1}{R} \frac{i_C}{C} = \frac{i_C}{RC} = \frac{i_{in} - i_R}{RC} = \frac{i_{in}}{RC} - \frac{i_R}{RC}$$

$$\frac{d}{dt} \begin{bmatrix} i_R \\ e_{out} \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_R \\ e_{out} \end{bmatrix} + \begin{bmatrix} \frac{1}{RC} \\ \frac{1}{C} \end{bmatrix} i_{in}$$

Steady state of this system will be $e_C(t \gg \tau) = e_{out}$. since $I = 1A$ and $R = 1\Omega$, $e_{out} = 1V$.

time constant τ can be denoted by the product of C and R , which is $1 [F] \cdot 1 [\Omega] = 1 [s]$.

